

Mutual Fund Analytics Capstone Project

Final Project Report

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1. Executive Summary

The Mutual Fund Analytics Capstone Project aims to build a complete analytics platform for analysing mutual fund performance and investor behaviour. The project covers the complete data analytics lifecycle including data ingestion, cleaning, exploratory data analysis, performance metrics, dashboard creation and advanced risk analytics.

Python was used for data processing and statistical analysis, SQLite for structured storage, and Power BI for interactive dashboard development. Advanced analytics techniques such as Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Cohort Analysis and a Fund Recommendation Engine were implemented.

2. Project Overview

The project focuses on analysing mutual fund datasets to help investors understand fund performance, investment risk and market trends.

The platform provides:

- Automated data processing
- Performance metrics

- Investor analytics
 - Interactive dashboards
 - Risk assessment
 - Recommendation system
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3. Problem Statement

Mutual fund investors often struggle to evaluate thousands of investment schemes because financial data is distributed across multiple datasets.

The objective of this project is to create an analytics platform capable of transforming raw data into meaningful insights through automated processing and visualization.

4. Objectives

- Build an automated ETL pipeline.
- Clean and validate datasets.
- Store processed data efficiently.
- Perform exploratory data analysis.
- Calculate financial performance metrics.
- Develop an interactive Power BI dashboard.

- Perform advanced risk analytics.
 - Recommend suitable funds based on investor risk profile.
-

5. Dataset Description

The project uses four datasets.

Fund Master

Purpose:

Contains information about every mutual fund scheme.

Columns include:

- AMFI Code
 - Scheme Name
 - Fund House
 - Category
 - Plan
 - Risk Grade
-

NAV History

Purpose:

Stores historical Net Asset Value data.

Columns:

- AMFI Code
 - Date
 - NAV
-

Scheme Performance

Purpose:

Contains risk and return statistics.

Columns:

- Alpha
 - Beta
 - Sharpe Ratio
 - Sortino Ratio
 - Expense Ratio
 - AUM
 - Maximum Drawdown
-

Investor Transactions

Purpose:

Contains investor transaction records.

Columns:

- Investor ID
 - Gender
 - State
 - Transaction Type
 - Amount
 - Transaction Date
-

6. Project Architecture

Workflow:

Raw CSV Files



Data Ingestion



Data Cleaning



SQLite Database



Exploratory Data Analysis



Performance Analytics



Power BI Dashboard



Advanced Analytics

7. Technology Stack

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib

Database

- SQLite

Visualization

- Power BI

Version Control

- Git & GitHub
-

8. Day 1 – Data Ingestion

Completed Tasks

- Project folder setup
- Virtual environment creation
- Package installation
- Data ingestion scripts
- Live NAV fetch

Outcome

Successfully loaded raw datasets into the project.

9. Day 2 – Data Cleaning

Cleaning Steps

- Removed duplicate records
- Handled missing values
- Converted date columns
- Standardized transaction types
- Validated expense ratio
- Saved cleaned datasets

Outcome

Generated high-quality processed datasets ready for analysis.

10. Day 3 – Exploratory Data Analysis

Performed analyses on:

- Fund categories
- Risk grades
- Investor demographics
- State-wise investments
- Transaction trends

Key Findings

- Equity funds dominate the dataset.
 - SIP transactions contribute a significant share of investments.
 - Large metropolitan states show higher investment volumes.
-

11. Day 4 – Performance Analytics

Metrics Calculated

- Annual Return
- Annual Volatility

- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Expense Ratio
- AUM

Output

Generated fund_scorecard.csv for detailed comparison.

12. Day 5 – Power BI Dashboard

Dashboard Pages

1. Industry Overview
2. Fund Performance
3. Investor Analytics
4. SIP & Market Trends

Dashboard Features

- KPI Cards
- Interactive Slicers
- Drill-down Visuals

- Dynamic Charts
-

13. Day 6 – Advanced Analytics

Implemented

- Historical VaR (95%)
- Conditional VaR
- Rolling Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Fund Recommendation Engine

Generated Files

- var_cvar_report.csv
 - rolling_sharpe_chart.png
 - Advanced_Analytics.ipynb
 - recommender.py
-

14. Project Structure

Include your project folder structure exactly as submitted.

15. Challenges Faced

- File path errors
- Git configuration issues
- Power BI measure errors
- Data type mismatches
- Missing folders
- Dataset validation

Solutions

Resolved using dynamic paths, data validation and structured project organization.

16. Results & Key Insights

- High Sharpe Ratio funds demonstrated better risk-adjusted performance.
- Lower expense ratio funds generally achieved stronger long-term returns.
- SIP investments showed consistent growth.
- Power BI dashboards enabled effective visualization of investment trends.
- Recommendation engine successfully suggested funds based on investor risk appetite.

17. Future Enhancements

- Live AMFI API integration.
- Streamlit web application.
- Machine Learning recommendation engine.
- Portfolio optimization.
- Automated email reporting.

18. Conclusion

The Mutual Fund Analytics platform successfully transformed raw financial data into meaningful business insights using Python, SQLite and Power BI. The project demonstrates the complete analytics workflow from ETL to dashboard development and advanced financial risk analysis.

19. References

- AMFI India
- Python Documentation
- Pandas Documentation
- NumPy Documentation

- [Matplotlib Documentation](#)
 - [Power BI Documentation](#)
 - [SQLite Documentation](#)
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